

What atlas is -

Atlas is a model in work that is based off of the betting service PrizePicks.

PrizePicks is the largest daily fantasy sports (DFS) platform in North America, designed as a "skill-based" alternative to traditional sports betting. Unlike standard DFS sites where you compete against other players in salary-cap tournaments, PrizePicks lets you play directly "against the house" by making over/under predictions on professional athlete stats.

How the Gameplay Works

The core experience involves building a "Lineup" of 2 to 6 players from various sports.

Player Picks: You select a specific stat for a player (e.g., LeBron James points) and predict if they will get More or Less than the projection set by PrizePicks.

Pick Types:

Standard Projections: Typical over/under lines for common stats like points, rebounds, or touchdowns.

Demons & Goblins: "Demon" picks feature harder-to-hit, higher projections that offer much larger payouts (up to 2000x), while "Goblins" have lower projections that are easier to hit but yield smaller returns.

Entry Modes:

Power Play: Higher risk; all picks in your lineup must be correct to win.

Flex Play: Lower risk; you can still win a smaller payout even if one or two of your picks are incorrect.

Sports & Categories Covered

PrizePicks offers a massive variety of markets across 100+ sports and categories:

Major Leagues: NFL, NBA, MLB, NHL, and WNBA.

College Sports: CFB and CBB (availability varies by state).

Niche & Global Sports: Soccer, UFC, Tennis, and Golf.

Esports: Popular titles like CS2, League of Legends, and Dota 2.

Culture Picks: Predictions on non-sporting events like awards shows or reality TV.

PrizePicks

Legality & Availability

Because it is classified as a "game of skill" rather than gambling, PrizePicks is available in many regions where traditional sportsbooks are not.

Regional Reach: It currently operates in over 30 U.S. states (including California, Texas, and New York) and parts of Canada.

Platform: Accessible via the PrizePicks App on iOS/Android or through their official website.

Wikipedia

Financials & Rewards

Payouts: Can range from 2x your entry fee up to 2000x for a 6-pick lineup using "Demon" picks.

Promotions: New users often receive a deposit match (e.g., \$50 bonus for playing \$5). They also offer Free2Play contests where users can win real rewards without an entry fee.

This model is strictly designed to play the NBA stat lines. Our goal is to find the best possible 3-5leg outputs for a slip. I have separated the live runs into different categories (system - uses the kernel based off PrizePicks goblin, standard, and demon multipliers, winprob which is strictly probability picks, and then windfall which is a combo of system and winprob picks that give more of an edge multipliers). All of these have Goblin, Standard, and Demon picks to make the PrizePick multiplier winnable and also playable to actually win value.

Everything in this model has a purpose and I've tried my best to standardize a solid model that produces slips using a Monte Carlo. It fetches a live api from PrizePicks converts from there into csvs for for slips. The main focus is always to better the model using math and real basketball philosophy on game type situations. I have a live injury api pulled that removes the out or doubtful players from the slips immediately. The questionable players are designed to be shown up in the slips as questionable. This is to prevent players that shouldn't be in slips be in the at all from the start. Those legs can kill slips. The next piece of information that is important is the role allocator. It takes those injuries and disperses the injured players stats accordingly to the remaining players. If a star player is out the distribution is high with multiple stats and higher value stat lines distributed among the team. If it is a lower quality player or a player who does not play it is a minimal distribution or none at all. Some teams that are deep and have multiple high value players are not as affected by one key player out near as much where a team that might have their star player out might be greatly affected. That is the design.

Another important part of the model is things like minutes and usage. Those should and are supposed to play a key role in how the allocator views these players according to their value based on their stat lines. Also, the model pulls the spread of each game via api that creates the fragility of a leg. If there is a blowout possible the leg becomes unstable specifically the over line on the prop. The under should not be affected as much or at all by this because if the blowout is a real possibility then the stars usually have minutes cut or none existant in the second half or fourth quarter. That makes the under line much more appealing to those players. The opposite could be said by a backup bench guy, if the blowout is possible a bench guys stat line should have a greater chance of increasing. Just basic basketball thoughts and philosophy. That is what I'm trying to accomplish is this basic philosophy using math and code to give me the best slips for that days games. Everyday is a different set of games, injuries, and slips with its own unique set of challenges. If a live run is done at say 1pm it might look completely different by game time based on the new injury output closer to game time.